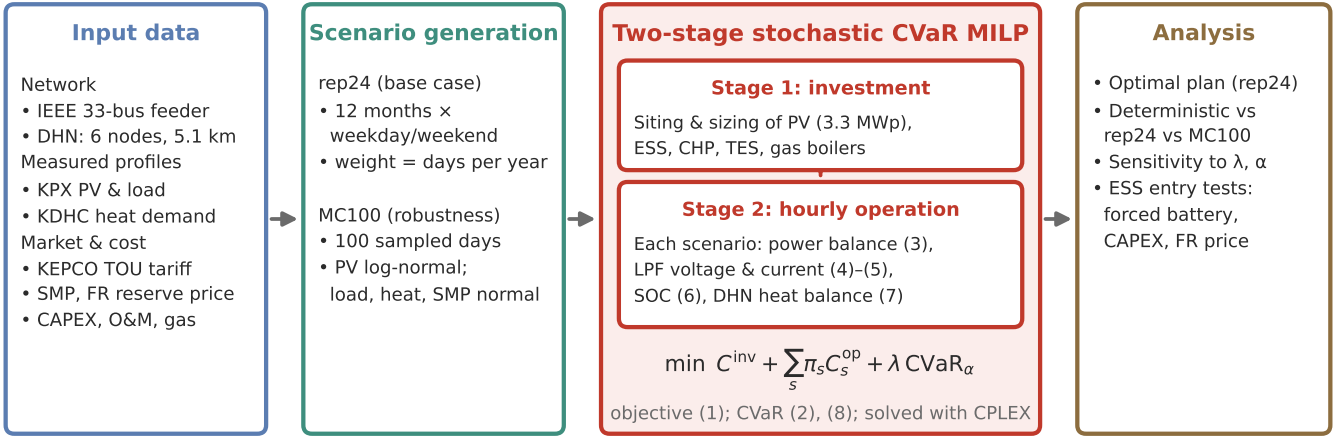
Input data
Network
• IEEE 33-bus feeder
• DHN: 6 nodes, 5.1 km
Measured profiles
• KPX PV & load
• KDHC heat demand
Market & cost
• KEPCO TOU tariff
• SMP, FR reserve price
• CAPEX, O&M, gas
Scenario generation
rep24 (base case)
• 12 months × weekday/weekend
• weight = days per year
MC100 (robustness)
• 100 sampled days
• PV log-normal; load, heat, SMP normal
Two-stage stochastic CVaR MILP
Stage 1: investment
Siting & sizing of PV (3.3 MWp), ESS, CHP, TES, gas boilers
Stage 2: hourly operation
Each scenario: power balance (3), LPF voltage & current (4)–(5), SOC (6), DHN heat balance (7)
min $C^{inv} + \sum_s \pi_s C_s^{op} + \lambda\,\mathrm{CVaR}_\alpha$
objective (1); CVaR (2), (8); solved with CPLEX
Analysis
• Optimal plan (rep24)
• Deterministic vs rep24 vs MC100
• Sensitivity to λ, α
• ESS entry tests: forced battery, CAPEX, FR price